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Web3 and AI Engineer · Daml on Canton · Solidity / EVM · Agentic LLM Systems

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Eight years in blockchain. I anchor what AI agents do onto chains, so nobody has to take the operator's word for it. On Polygon that is the activity trail. On Canton, in Daml, it is also what the agent was **refused**, which is the part a ledger normally keeps nothing of. I write the contracts and the agentic layer that drives them.

SMART CONTRACTS AND LEDGERS

- **Canton and Daml. 3rd place, Cantor8 “Build on Canton”**, London 2026, judged by a panel from across the Canton ecosystem. Package **kya-rails-mandate**, SDK 3.4.11, vetted on Canton DevNet. 111 Daml test scripts, 32 spending fences, 82 mutations that each break something real and require a named test to go red.
- **Solidity and EVM**. Smart contracts written and deployed on Polygon. At **Veritaport** (veritaport.co.uk), the evidentiary layer of a forensic system: records anchored on chain so a third party verifies integrity in seconds, with no per-user wallets. At **Matty's Place** (app.mattysplace.org.uk), every action an AI agent takes is anchored to Polygon, so the agent's activity trail is tamper-evident rather than self-reported.
- **Accepted by other people**. 3 pull requests merged into the **Canton Developer Hub** (#156, #160, #168). 4 filed into the **Cantor8 hackathon toolkit** (#7 to #10). Dev Fund proposal open under RFP 27, security monitoring and auditability.
- **Security findings**. 3 reports to OpenZeppelin's Canton libraries: **52 of 69 authorisation fences could be made vacuous with every test still passing**.

AGENTIC AI AND LLM SYSTEMS

- **Know Your AgenticAI**. A spend-limited wallet for an AI agent. The cap, the allow-list and the period limit are enforced inside a Daml choice, never in Python. Every attempt including every refusal becomes a hash-chained receipt, and the refusal itself commits to the ledger as a contract rather than a log line. Ships with an MCP server, so a model cannot overspend its mandate.
- **Production agents**. Tool use, planning and memory; multi-step orchestration; human-in-the-loop control; RAG and OCR pipelines. LangChain, LangGraph, CrewAI, AutoGen, MCP-style structured context. Hallucination, latency and cost trade-off design.
- **AI Code Quality Auditor**. Measures where AI coding agents fail across hallucination, complexity, duplication and security density. A controlled 600-measurement experiment across four agents, with SonarCloud integration. The core instrument of an MSc at Aston.

PUBLISHED SOFTWARE

- **knowyouragenticai-receipts** (PyPI). A receipt format with a written specification and 20 conformance vectors. Three independent implementations, in Python, JavaScript and Go, agree on all 20. MIT, zero dependencies. 241 downloads a month.
- **ai-code-quality-auditor** (PyPI). The MSc instrument, published as reusable research infrastructure. 472 downloads a month.

SELECTED DELIVERY

- **Eagles Den, Vorem**. Paid-subscriber analytics on Google Cloud Run. Five-model LLM workflow, Paystack billing, Telegram and WhatsApp delivery. Live subscribers and revenue, with ongoing ownership of uptime, latency and cost.
- **Matty's Place tenant hub, via MCKI Solutions** (app.mattysplace.org.uk). Enterprise HMO platform. 100+ tenants, multi-brand, role-based access architecture, SHA-256 audit chain, agent activity anchored to Polygon. Saved roughly 100 hours a week and £80,000 a year.
- **Nigeria tax calculator**. Retrieval and OCR reading bank statements and classifying transactions for tax. Deployed.

STACK

- Daml, Solidity, Python, TypeScript and JavaScript, SQL. FastAPI, Next.js, Docker, CI/CD, automated testing, PyPI packaging. Google Cloud Run, Azure, AWS, Vercel, Fly.io, Supabase, Snowflake. PyTorch, TensorFlow. STRIDE threat modelling, FAIR risk scoring.

EDUCATION

- **MSc Artificial Intelligence and Business Strategy**, Aston University, 2026. Dissertation: measuring the quality and safety of AI-generated code.

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